

Isaac Tunney

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EDUCATION

Postdoctoral in Mechanical Engineering / Robotics

2026

Development of a novel ROS-based passive landing gear for a tree-planting drone
Université de Sherbrooke

PhD in Mechanical Engineering (Fast-track entry from Masters) – GPA: 4.3/4.3

2021-2026

Project focused on aerial robotics
Université de Sherbrooke

Bachelor degree in Mechanical Engineering (Coop Program) – GPA: 4.01/4.3

2016-2020

Aeronautic Concentration
Université de Sherbrooke

KNOWLEDGE

Linguistic French (first language), English (oral 4.5/5, written 5/5)

Software

- Programming: C++, Python, MATLAB, Simulink, Arduino / Windows and Linux (Ubuntu)
- Design: AutoCAD, OnShape, SolidWorks
- Dynamics/Robotics: Motion Genesis, Gazebo Simulator, ROS, Simscape, WM2D
- Motion Tracking: Optitrack, MOCAP, Phantom High-Speed camera

Professional Development

- Advanced Dynamics Course (*Université de Sherbrooke*)
- Aerial Robotics Course on *Coursera* – online certificate
- C++ Course on *Udemy* – online certificate
- Canadian pilot certificate for basic drone operations – 40h+ of flight time with multirotors
- EASA pilot license for Iceland's A1/A3 drone category – 5h of flight time with multirotors

DOCTORAL PROJECT (2021-2026)

Objective: Develop a landing technology combining **friction-based shock absorbers** (hardware) and **reverse thrust** (software) to enable small drones to land rapidly and reliably in **extreme outdoor conditions**.

Key Contributions:

- Designed a novel landing gear using friction-based shock absorbers for **high-speed impacts** (<4 m/s)
- Developed custom landing algorithms in **C++ (ArduPilot)**, integrating reverse thrust to prevent flipping
- Sized and built a **high-power UAV** (thrust-to-weight ratio of 8) designed for speed and maneuverability
- Ran **500k+ dynamic simulations** (MATLAB) with a realistic 2D UAV model to predict landing success rates
- Designed **specialized feet** with **retractable spines** to enable landings on steep, ice surfaces like icebergs
- Demonstrated landings in **harsh real-world conditions**: Rooftops inclined up to 60° at descent speeds up to 3 m/s, moving vehicle at 120 km/h, boats at 30 km/h, icebergs & glaciers with slopes up to 60° in Iceland

Dissemination:

- Public video demonstrating a drone landing on a high-speed vehicle: <https://youtu.be/tTUVr1Ogag0>
- Public video demonstrating a drone landing on rooftops: https://youtu.be/tG1K_63q00Y
- Public video demonstrating a drone landing on a fast canoe: <https://youtu.be/E-ns-MxMJvU>

Doctoral Publications and Presentations:

- *Journal paper (Submitted)*: I. Tunney, J. Bass and A. L. Desbiens, "The Ice UAV: Landing on Steep Ice Using Spiny Feet, Dissipative Landing Gear and Reverse Thrust", in Transactions on Field Robotics, 2026
- *Seminar at Carleton University*: I. Tunney, "Friction Shock Absorbers and Reverse Thrust for Fast UAV Landing on Icebergs and Glaciers", NSERC CREATE UTILI Workshop, 2025
- *Journal paper*: I. Tunney, J. Bass and A. L. Desbiens, "Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on High-Speed Vehicles" Journal of Field Robotics, 2025
- *Journal paper & conference*: J. Bass, I. Tunney and A. L. Desbiens, "Adaptative Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on Inclined Surfaces," in IEEE RA-L, 2022

POSTDOCTORAL PROJECT (2026)

Objective: Develop and deliver a passive **self-levelling landing gear** for a 30-kg UAV used for tree-planting operations on rough, inclined forestry terrain within a 6-month timeframe.

Key Contributions:

- **Authored** the successful proposal for **SPADE Open Call #2** (Horizon Europe Program), securing **\$100k CAD** through competitive international selection.
- **Technical lead** for a 4-person team covering PCB design, ROS integration, embedded software, mechanics, manufacturing, and communications with the client (SPADE/NIBIO) in Norway.
- **Designed all main components of the novel landing gear**, including rack-and-pinion transmission, servo-actuated friction brakes, encoder feedback, and the carbon-fiber, aluminum, and 3D-printed structure.
- **Developed multi-level control algorithms** to adjust leg friction for height and tilt control of the 4-legged UAV (control scheme for over-constrained systems).

OTHER ACADEMIC PROJECTS

Research Project (in parallel with PhD) – Novel grippers for acrobatic robots

2023-2024

- Technical lead on a 1.5-year project funded by a large U.S. robotics company (\$100k+ CAD funding)
- Designed and built a novel robotic hand enabling an acrobatic robot to release itself from a swinging bar and catch itself onto a new bar (>30G at impact) less than 1 second later, repetitively
- Oversaw the global project's heading, recruited and supervised a Master's student on the project
- Ran dynamic simulations (using *WM2D*) to understand key challenges; iterated over 50 mechanisms
- Delivered a final functional prototype: single solenoid to open the hand, spring-driven passive resetting of the gripper's fingers and thumb under 0.75s, high mechanical advantage to release under high loads

Senior design project – AeroStrabe: A Single-Seat Quadcopter-Type Vehicle Simulator for Urban Air Mobility

2019-2020

- Developed a dynamic model of the vehicle in Python to realistically simulate flight behavior
- Programmed the motion system (4 actuators + 1 motor) to generate realistic perceived motion (C++)
- Integrated an Oculus VR headset with a custom virtual environment built with *Unreal Engine*
- Designed a real-time software architecture to synchronize all simulator submodules
- Secured a \$10,000 USD grant from *Epic Games* (Epic MegaGrants) for innovation in virtual reality
- Project completed by a group of 6 students in 12 months with a total budget of \$50,000 CAD
- Project video: <https://www.facebook.com/watch/?v=3886459841378560>

5th semester integration project: Fully motorized heightening system to help attendants in care centers to raise their patients with reduced mobility

2018

4th semester integration project: Rocket flight simulation on Mars and geometric optimization to maximize distance traveled (in MATLAB)

2018

WORK EXPERIENCE

- Walt Disney Imagineer, Disney Research (California, US) – 4-month internship** 2023
- Developed first prototype of an acrobatic robot performing mid-air maneuvers
 - Programmed embedded software in Python on a Raspberry Pi CM4 with multithreaded architecture
 - Fully designed and built a custom 2-DOF robot using metal laser cutting and 3D printing
 - Contributed to electronics system design: radio system, power module, motors, sensors, and wiring
 - Simulated robot dynamics and control using Simscape Multibody (MATLAB) prior to implementation
 - Integrated and fused sensor data (IMU, motion capture) for real-time tracking
 - Created the global system architecture: real-time comms. between robot, pendulums and MOCAP
- Software specialist in flight aerodynamics, CAE (Montreal, Qc) – 4-month internship** 2019
- Developed and implemented flight dynamics programs in C++ for Boeing B747 Simulator
 - Automated flight database updates with Python, reducing task time from >35h to <1min
- Aerial Robotics Designer, Outreach Robotics / Createk (Sherbrooke, Qc) – 4-month internship** 2019
- Designed 4 core mechanical & robotic components, from concept to prototype: main grippers to grasp branches, orientation-locking serrated joint, emergency release, and full structural assembly
 - Contributed to the *DeLeaves* product now commercialized by *Outreach Robotics*
 - Rapidly evaluated designs through iterative prototyping (3D printing, laser cutting, machining)
- Staff Coordinator, Department of Production, ArcelorMittal Produits Longs Canada (Contrecoeur, Qc) – 4-month internship** 2018
- Managed day-to-day operations of the steel casting department, overseeing ~30 employees
 - Handled equipment failures, enforced safety procedures, responded to injuries, etc.
- Mechanical Engineering Intern, Cycles Devinci (Chicoutimi, Qc) – 4-month internship** 2017
- Designed and built an automated corrosion bath to test anti-corrosion products on bike parts
 - Improved a test bench used to compare stiffness between different bicycle frames

RECOGNITIONS

- Recipient of SPADE Open Call #2 (Horizon Europe), awarded \$100k in project funding 2025
- Recipient of the CAMAQ scholarship for the next generation in aeronautics (small money prize) 2024
- First place at the CIADI Capstone Project Competition (Concordia University); Money prize and invitation to the 2024 ICAO Symposium on Advanced Air Mobility. 2024
- UTILI Scholarship / member of UTILI program (<https://carleton.ca/utili/isaac-tunney/>) 2023
- Certificates of academic excellence (top 10%) from the Université de Sherbrooke 2019, 2020
- NSERC and FRQNT scholarships for research internships at the Createk Design Lab 2019

PERSONAL PROJECTS AND INTERESTS

- Simulation of two drones (using Ardupilot) controlled simultaneously with ROS in the Gazebo Simulator
- RRT motion planning algorithm and minimum snap trajectory generation for a quadcopter in MATLAB
- Omnidirectional rolling robot to play and interact with dogs, launch treats and throw balls (ongoing)
- Small-distance radar with a graphic interface (programmed in Java)
- Programs in C++ and Python, including graphical animations and a financial web scraper
- **Personal interests:** rock climbing, running, tennis, coffee / latte art design, and everything related to robots!